

Playwright Complete Guide

Microsoft Playwright is an open-source end-to-end testing framework for web applications.

Overview

Supports Chromium, Firefox, and WebKit. Enables reliable browser automation and testing.

Key Features

Cross-browser testing, auto-waiting, screenshots, video recording, tracing, API testing, parallel execution.

Installation

`npm init playwright@latest` or `npm install -D @playwright/test`.

Core Concepts

Browsers, contexts, pages, locators, assertions, fixtures, and test runners.

Locators

`getByRole`, `getByText`, `getByLabel`, CSS selectors, XPath selectors.

Assertions

`expect(locator).toBeVisible()`, `toHaveText()`, `toHaveURL()`, and more.

API Testing

Use Playwright request context to validate REST APIs.

Reporting

HTML reports, traces, screenshots, videos, CI/CD integration.

Best Practices

Use resilient locators, isolate tests, avoid hard waits, use page objects where appropriate.

Sample Test

```
test('login', async ({ page }) => { await page.goto(url); });
```